

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12410816
Kind Code	B1
Date of Patent	September 09, 2025
Inventor(s)	Conti; Michael Paul et al.

Bladeless ceiling fan

Abstract

A ceiling fan comprising a stationary inner housing defining a first interior, a stationary outer housing that circumscribes at least a portion of the stationary inner housing and defines a second interior, a motor assembly located within the first interior and having a rotor, and a plurality of blades extending between corresponding roots and tips, with the roots coupled to the rotor and the tips extending toward to the outer housing. The tips are coupled to a tip housing that can at least partially be received within a portion of the outer housing. A flow channel is formed between the stationary inner housing and the stationary outer housing.

Inventors: Conti; Michael Paul (Memphis, TN), Mason; Douglas Troy (Horn Lake, MS), Botkin; Charles William (Cordova, TN)

Applicant: Hunter Fan Company (Memphis, TN)

Family ID: 96950415

Assignee: Hunter Fan Company (Memphis, TN)

Appl. No.: 19/188407

Filed: April 24, 2025

Publication Classification

Int. Cl.: F04D25/08 (20060101); F04D19/00 (20060101); F04D29/00 (20060101); F04D29/38 (20060101); F04D29/54 (20060101)

U.S. Cl.:

CPC F04D29/542 (20130101); F04D19/002 (20130101); F04D25/088 (20130101); F04D29/005 (20130101); F04D29/384 (20130101);

Field of Classification Search

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9719525	12/2016	Cunnane	N/A	F04D 29/441
10487852	12/2018	Avedon	N/A	F04D 29/601
11022127	12/2020	Hiner	N/A	N/A
11480193	12/2021	Whitmire	N/A	N/A
12085083	12/2023	Botkin	N/A	F04D 29/30
12180976	12/2023	Arens	N/A	N/A
2017/0370363	12/2016	Avedon	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
212615465	12/2020	CN	N/A
217152342	12/2021	CN	N/A
220622218	12/2023	CN	N/A
I509159	12/2014	TW	N/A
2023272915	12/2022	WO	N/A

OTHER PUBLICATIONS

Exhale Fans, “Bladeless Ceiling Fans”, <https://exhalefans.com/product/bladeless-ceiling-fans/> Feb. 14, 2025. cited by applicant

Home Depot, “19.7 LED Indoor Black Bladeless Low Profile Ceiling Fan Flush Mount Smart App Remote Control Dimmable Lighting”, https://www.homedepot.com/p/Lasaideal-19-7-in-LED-Indoor-Black-Bladeless-Low-Profile-Ceiling-Fan-Flush-Mount-Smart-App-Remote-Control-Dimmable-Lighting-HDL1340C789BK/331816277?emt_ppr_vfpd Feb. 14, 2025. cited by applicant

Target, “Bella Depot Bladeless Ceiling Fan with Light and Remote”, <https://www.target.com/p/bella-depot-bladeless-ceiling-fan-with-light-and-remote/-/A-1001379792#Ink=sametab> Feb. 14, 2025. cited by applicant

Primary Examiner: Hasan; Sabbir

Attorney, Agent or Firm: McGarry Bair PC

Background/Summary

TECHNICAL FIELD

(1) This disclosure is directed to a ceiling fan, and more specifically, to a bladeless ceiling fan.

BACKGROUND

(2) Ceiling fans are traditionally suspended from a structure for moving a volume of air about an area, such as a room. The ceiling fan uses a motor that drives a set of fan blades to rotate and circulate air. Rotating blades drive air, often to achieve a cooling effect, but will also vibrate and emit noise while the ceiling fan is operational.

BRIEF DESCRIPTION

(3) In one aspect, the disclosure relates to a ceiling fan comprising a stationary inner housing defining a first interior and having a circumferential rotor channel, a stationary outer housing that circumscribes at least a portion of the stationary inner housing to define a flow channel therebetween and defines a second interior having a circumferential tip channel, a motor assembly located within the first interior and having a rotor aligned with the rotor channel, a plurality of blades extending between corresponding roots and tips with the roots coupled to the rotor, and a tip housing coupled to the blade tips and at least partially received within the tip channel.

(4) In another aspect, the disclosure relates to a ceiling fan comprising a stationary housing defining a first interior and having a circumferential rotor channel, a motor assembly located within the first interior and having a rotor aligned with the rotor channel, a plurality of blades extending between corresponding roots and tips, with the roots coupled to the rotor; and a tip housing coupled to the blade tips.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

(2) FIG. 1 is a perspective view of an exemplary bladeless ceiling fan.

(3) FIG. 2 is a cross-sectional view of the exemplary ceiling fan of FIG. 1 and illustrating a tip housing.

(4) FIG. 3 is an enlarged view of the tip housing.

(5) FIG. 4 is a perspective view of another exemplary ceiling fan system.

DETAILED DESCRIPTION

(6) The disclosure relates to a bladeless ceiling fan, a corresponding kit, along with the components forming the assembly or kit. The resulting ceiling fan can be used, for example, in residential and commercial applications. Such applications can be indoors, outdoors, or both. While this description is primarily directed toward a residential ceiling fan, it is also applicable to any environment utilizing fans or for cooling areas utilizing air movement.

(7) Bladeless ceiling fans are a subset of ceiling fans. Bladeless fans do have blades. However, unlike normal ceiling fans, bladeless fans completely or partially hide the blades, especially the blade tips, to make the blades less visible or completely invisible to the user. To hide the blade tips, the blade tips can reside within a housing or shroud, which is typically a circumferential housing defining a non-observable volume through which the blade tips rotationally traverse. The close presence of the housing to the blade tips creates a substantial amount of aerodynamic turbulence that can increase the torque demands on the motor and/or reduce the efficiency of the air flow through the fan. A benefit of the bladeless fan as described is that it reduces the aerodynamic turbulence.

(8) As used herein, the term “set” or a “set” of elements can be any number of elements, including only one. All directional references (e.g., radial, axial, proximal, distal, upper, lower, upward, downward, left, right, lateral, front, back, top, bottom, above, below, vertical, horizontal, clockwise, counterclockwise, upstream, downstream, forward, aft, etc.) are only used for identification purposes to aid the reader's understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of aspects of the disclosure described herein. Connection references (e.g., attached, coupled, connected, and joined) are to be construed broadly and can include intermediate members between a collection of elements and relative movement between elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and in fixed relation to one another. The exemplary drawings are for purposes of illustration only and the dimensions, positions, order and

relative sizes reflected in the drawings attached hereto can vary.

(9) FIG. 1 illustrates a ceiling fan **10** in the form of a bladeless ceiling fan extending between a lower end **12** and an upper end **14**, configured so the upper end **14** is adjacent a structure, such as a ceiling, when mounted. The ceiling fan **10** includes a stationary housing assembly **15**, a set of blades **20**, and an optional light assembly **22**. The stationary housing assembly **15** includes a stationary inner housing **24** and a stationary outer housing **26** circumscribing at least a portion of the inner housing **24**. Together, the inner and outer housings **24**, **26** define an outlet **28**, and an inlet **30** and a flow channel **32** therebetween. The flow channel **32** generally has an annular shape and extends between the lower end **12** and the upper end **14** of the ceiling fan **10**. In the illustrated example, the inlet **30** of the flow channel **32** is located at the upper end **14** and can draw in air that moves through the flow channel **32** to be expelled from the outlet **28** at the lower end **12**. The ceiling fan **10** can be reversible such that air is drawn in at the lower end **12** and expelled aft as illustrated by dotted arrow **33**, which reverses the function of the outlet **28** and inlet **30**.

(10) The set of blades **20** can be located between the inner and outer housings **24**, **26** and rotated about a rotational axis **38**. In some examples, as illustrated, one or both of the inner housing **24** and the outer housing **26** can be concentric and symmetric about the rotational axis **38**.

(11) The light assembly **22** includes one or more lights **40** carried by the stationary housing assembly **15** at or adjacent to the lower end **12**. The lights **40**, as illustrated, are carried by the inner housing **24**. However, the lights **40** could just as easily be carried by the outer housing **26** or both. The lights **40** can, in some examples, be a plurality of light emitting diodes (LEDs) that couple to a light housing (not shown) located within the first interior **34**, second interior **36**, or both. Other illumination sources, like traditional light bulbs may also be used instead of or in combination with the LEDs. The light assembly **22** can be supplied power by passing electricity and data lines through one or both of the inner or outer housings **24**, **26**. The light assembly **22** can also enclose mechanical or digital switches for turning on/off/dimming of the lights **40**. In a variation where the lights **40** are not included, a cover (not shown), including an opaque cover, can be used instead of the lights **40**.

(12) Turning to FIG. 2, a cross-sectional view of the ceiling fan **10** is shown mounted to a structure **42**. It should be noted that the section seen in FIG. 2 is generally a vertically-oriented section plane, except for the blades, which are sectioned along their length, not in a vertical plane. A set of struts **84** extends between the outer housing **26** and the inner housing **24** and are configured to support and secure the outer housing **26** relative to the inner housing **24**. The set of struts **84** can advantageously be vanes designed to direct the flow channel **32**. The vanes can be a simple deflector, such as a louvre, or can have an airfoil cross section to further alter the airflow. In the illustrated example, the set of struts **84** are located above the set of blades **20**; however, other configurations are contemplated, such as the struts **84** being located below the blades **20**, or both above and below the blades. One or more of the struts **84** can be hollow or have an internal passage through which electrical power or data cables can pass, which is especially useful when the light assembly **22** is located on the outer housing **26**.

(13) The stationary inner housing **24** defines a first interior **34** that is at least partially hollow and can be used to house and visually obscure fan components used for mounting and/or fan operation. The stationary outer housing **26** defines a second interior **36**, which can also be at least partially hollow and used in the same manner as the first interior. Additionally, when a light kit is mounted to the stationary outer housing **26**, the second interior can be used for located the illumination sources, such as the LEDs, and/or passing the power or data cables.

(14) The inner housing **24** defines the upper end **14** of the ceiling fan **10** that, when mounted, is disposed at or adjacent to the structure **42**. The inner housing **24** extends from the upper end **14**, along the rotational axis **38**, to the lower end **12**. The inner housing **24** has an outer radial surface or first surface **44** bounding the first interior **34**.

(15) As illustrated, the first surface **44** forms at least a portion of the flow channel **32** and can be a

simple contoured surface or a more complex contoured surface. As shown, the first surface **44** is a more complex contoured surface in that it is similar to a parabola that is truncated at the lower end **12**. Whether or not the first surface **44** is parabolic, it can taper such that first surface **44** is wider at the upper end **14** and narrows toward the lower end **12**. A truncated cone or multiple conic sections with decreasing diameters could yield a similar shape.

(16) The inner housing **24** comprises an upper portion **45** axially spaced from a lower portion **46** by a circumferential rotor channel **48**. As such, the upper portion **45** can be the portion housing parts for mounting the fan **10** and the lower portion **46** can, for example, be the portion for housing switches or wires for the ceiling fan **10** and the light assembly **22**. The rotor channel **48** can be a gap between the upper and lower portions **45**, **46** that is at least partially defined by the first interior **34**.

(17) The stationary outer housing **26** is radially spaced from the stationary inner housing **24** and defines at least a portion of the flow channel **32**. That is, the radial gap **78** extending between an outer radial surface or second surface **82** of the outer housing **26** and the first surface **44** of the stationary inner housing **24** can form the flow channel **32**.

(18) The outer housing **26** includes a circumferential tip channel **80** formed in the second surface **82** facing the radial gap **78** and defined by the second interior **36**. The tip channel **80** can at least partially overlap axially with the rotor channel **48** with respect to the rotational axis **38**. However, the axial overlap, if any, is a function of the blade shape.

(19) The second surface **82**, as illustrated, can be a contoured second surface comprising rounded outer radial surfaces, where the outer housing **26** can have a toroidal shape. As such, the contoured second surface **82** forms at least a portion of the flow channel **32**. The tip channel **80** can be located within the contoured second surface **82** and the outer surface **76** of the tip housing **72** can be shaped to follow a contour of the contoured second surface **82**.

(20) The ceiling fan **10** can include a mounting assembly **50** for mounting to the structure **42** and a motor assembly **52**. The structure **42** can be a ceiling, for example, from which the ceiling fan **10** suspends. It should be understood that the structure **42** is schematically shown by way of example only and can include any suitable building, structure, home, business, or other environment wherein moving air with a ceiling fan is suitable or desirable.

(21) The mounting assembly **50** is schematically shown and can, for example, include a mounting bracket **54** and a downrod **56**, such that the mounting bracket **54** is the primary coupling with the structure **42**. The downrod **56**, at one end, couples to the mounting bracket **54** and, at the other end, couples to the motor assembly **52**.

(22) The motor assembly **52** includes a motor **60** comprising a rotor **62** and a stator **63**, which can be partially or fully enclosed by a motor housing **64**. A non-rotating motor shaft **65** extends within the motor housing **64** to support the stator **63** and passes through the upper portion **45** of the inner housing **24** to connect to the downrod **56**. The motor shaft **65** also extends beyond the rotor **62** and stator **63** to couple together the upper and lower portions **45**, **46** of the inner housing **24**. Thus, the motor shaft functions as both the structure about which the rotor **62** rotates and the structure connecting the upper and lower portions **45**, **46** of the inner housing **24**. In this way, the upper and lower portions **45**, **46** of the inner housing can be spaced from each other to form the rotor channel **48**.

(23) The motor **60** is an external rotor motor, where at least a portion of the rotor **62** circumscribes or is radially exterior of the stator **63**, and the stator **63** is non-rotationally affixed to the motor shaft **65**. The rotor **62** is rotationally coupled to the non-rotating motor shaft **65**, such as by the use of one or more bearing assemblies **59**.

(24) The motor housing **64** and rotor **62** are arranged and coupled such that at least a portion of the motor housing **64** can rotate about the non-rotating motor shaft **65**. For example, the motor housing **64** can have a seat in which the rotor sits and is operationally secured to the motor housing. In such a configuration, the motor housing **64** could be molded plastic or formed metal that defines a seat

in which the rotor is pressed-fitted or mechanically attached.

(25) The rotatable motor housing **64** sits within the circumferential rotor channel **48**, forming an outer shell for the motor assembly **52** between the upper and lower portions **45, 46**.

Advantageously, the motor housing is shaped such that its exterior appears as a continuation of the first surface **44**. The rotatable motor housing **64** includes blade hubs **66** for coupling the set of blades **20** to the motor assembly **52**. The blade hubs can be any suitable type of blade hub in the industry. An example of which includes a snap-fit connector where the root of the blade is snap-fitted into the hub. The type of blade hub and connector is not material.

(26) The set of blades **20** include one or more blades that connect to the hubs **66** and can, together with the rotatable motor housing **64**, be rotated about the rotational axis **38**. The blades **20** radially extend between corresponding roots **67** and tips **68** such that the roots **67** couple to the hubs **66** and the tips **68** extend outward toward the outer housing **26**. The blades **20** have corresponding leading edges **70** and trailing edges **71** axially extending from the roots **67** to the tips **68** with respect to the rotational axis **38**. A tip housing **72** can extend circumferentially about the axis **38** and couple, operably and/or physically, together the tips **68** of the blades **20**.

(27) Referring to FIG. **3**, a portion of FIG. **2** is enlarged such that the tip housing **72** is shown in greater detail where it can comprise, or be in the form of, a ring that circumscribes and connects the set of blades **20** at their corresponding tips **68**. The tip housing **72** includes an inner surface **74** facing the inner housing **24** and an outer surface **76** facing the outer housing **26**. In some examples, as illustrated, the tip housing **72** can extend fully along the blade tips **68**. Additionally, or alternatively, the tip housing **72** can extend along at least a portion of the leading edge **69**, the trailing edge **70**, or a combination thereof. As such, the inner surface **74** of the tip housing **72** can be angled and extend toward the inner housing **24**.

(28) The blades tips **68** and tip housing **72** are arranged to align with the tip channel **80** and can at least partially receive the tip housing **72**. As such, at least a portion of the tip housing **72** can be received in the second interior **36** of the outer housing **26**. The tip channel **80** can be complementarily shaped and sized to receive the tip housing **72** such that the tip housing **72** substantially closes the tip channel **80**, as illustrated, except for a clearance gap **81** at the upper and lower ends of the tip housing **72**. The clearance gap **81** is sufficient to prevent the tip housing from rubbing against the outer housing **26** during rotation of the blades. While not shown, if desired, a low-friction, wear-resistance material could close off the clearance gap **81**. Alternatively, instead of the low-friction, wear-resistance material, bristles could close off the gap.

(29) In this configuration, the tip housing **72** effectively closes the tip channel **80**, except for the gap **81**. The continuous and smooth surface of the tip housing faces the air within the second interior **36**, instead of the discrete blade tips. The continuous and smooth surface of the tip housing **72** interacting with the air in the second interior generates substantially less turbulence as compared to the blade tips without the tip housing. The reduction of turbulence at the interface of the tip housing **72** and the outer housing **26** improves the flow of air through the flow channel **32**, which greatly improves the efficiency and effectiveness of the fan.

(30) Another way one can think of this is that without the tip housing **72**, the interface of the blade tips rotating through the tip channel **80** would create a zone of turbulence at the interface around the circumference, with greater turbulence being at the current location of the blade tips. The air flowing through the flow channel **32** would encounter this zone of turbulence and disrupt the air flow, even extending turbulence out further into the flow channel **32**. This zone of turbulence is created even when the inner surface of the outer housing is closed and the blade tips are adjacent the inner surface. By wrapping the blade tips in the tip housing **72** and rotating the blade housing at least partially within the tip channel, this zone of turbulence is greatly diminished, thereby reducing resistance of air flow through the flow channel and improving the efficiency of the fan.

(31) As currently understood, the generation of the turbulence will be described, with it not being limiting as understanding may change. The current understanding is based on the phenomenon that

fluid in contact with a surface moves at the same speed as the surface. For example, in a pipe, the fluid in contact with the stationary pipe wall is stationary, even while the fluidly is flowing through the pipe, which creates a boundary layer between the wall and the fluid, with large speed gradients from the wall toward the middle of the pipe. In the context of the ceiling fan, when the set of blades **20** are rotatably driven by the motor **60**, the air is pushed by the blades **20** and moved at the angular velocity of the blades **20**, which increase as a function of the radial distance from the axis of rotation. Thus, the air in contact with the blade has the same angular velocity as the blade along the radial length of the blade. For example, the angular velocity and thus the speed of the air in contact with the blade increases from **32a**, **32b**, to **32c**. However, the air in contact with the outer housing, which is stationary, has an angular velocity of zero, forming a boundary layer of air at the interface between the outer housing and the tip housing **72**, where the angular velocity is maximum. The boundary layer would create a certain degree of turbulence.

(32) Without the tip housing **72**, the blade tips would periodically create a disruption in the boundary layer and temporarily increase the turbulence at the location of the blade tip. It is believed that the receipt of the tip housing **72** in the tip channel **80** eliminates the space between the tips **68** and the stationary outer housing **26** that would otherwise form large transient speed gradients and turbulence in the boundary layer as the blade tips pass. The tip housing **72**, being continuous, presents a constant interface with the outer housing, which yields a constant boundary layer for a fixed rotational speed of the blades.

(33) This is probably best understood by looking at one point on the outer housing **26** while the tip housing **72** rotates by as compared to discrete blade tips rotating by. In the case with the tip housing, the air speed gradient is always the same since the continuous tip housing **72** presents a continuous interface with the outer housing. However, in the case of the blade tips, without the tip housing, the interface is variable. When a blade tip is at the location, the speed gradient is at its greatest. However, the blade tip will rotate out of the location, which will now see the space or void between blade tips. The air at the blade tip is much faster than the air between the blade tips. The spot on the outer housing then sees a period increase in air speed gradient as the blade tip appears followed by a decrease in the air speed gradient as the space between the blade tips appears. This cycle repeats as the blade tips rotate past the location. This varying of the air speed gradient creates a lot of turbulence as compared to the continuous interface presented by the tip housing **72**.

(34) FIG. **4** illustrates the ceiling fan **10** comprising an air splitter **86** fluidly coupled to the flow channel **32**. The air splitter **86** is in the form of a ring that extends circumferentially about the rotational axis **38** and is configured to interact with air moving through the flow channel **32**. The air splitter **86** can be one or more air splitters **86** located at the inlet **30**. In the illustrated example, the air splitter **86** is secured in place via struts **84**. Alternatively, it is contemplated that the air splitter **86** be coupled to edges of the set of blades **20**, for example, the trailing edges **71**.

(35) In addition to the benefits already described, enclosing blades within the inner and outer housings provide a smoother, quieter, and more efficient ceiling fan, improving consumer satisfaction. The ceiling fans as described herein provide for facilitating installation, and provide for a ceiling fan without any exposed hardware. Additionally, the tip channel provides for reduced blade sag, which can provide for improved ceiling fan lifetime and efficiency. This can further provide a quieter operation and an improved overall consumer experience.

(36) To the extent not already described, the different features and structures of the various features can be used in combination as desired. That one feature is not illustrated in all of the aspects of the disclosure is not meant to be construed that it cannot be but is done for brevity of description. Thus, the various features of the different aspects described herein can be mixed and matched as desired to form new features or aspects thereof, whether or not the new aspects or features are expressly described. All combinations or permutations of features described herein are covered by this disclosure.

(37) This written description uses examples to detail the aspects described herein, including the

best mode, and to enable any person skilled in the art to practice the aspects described herein, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the aspects described herein are defined by the claims, and can include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

(38) Further aspects are provided by the subject matter of the following clauses:

(39) A ceiling fan comprising: a stationary inner housing defining a first interior and having a circumferential rotor channel; a stationary outer housing, circumscribing at least a portion of the stationary inner housing to define a flow channel therebetween, and defining a second interior having a circumferential tip channel; a motor assembly located within the first interior and having a rotor aligned with the rotor channel; a plurality of blades extending between corresponding roots and tips, with the roots coupled to the rotor; and a tip housing coupled to the tips of the blades and at least partially received within the tip channel.

(40) The ceiling fan of claim any preceding clause, wherein the tip housing extends fully along the tips.

(41) The ceiling fan of claim any preceding clause, wherein the blades have corresponding leading edges and trailing edges extending from the roots to the tips, and the tip housing extends along at least a portion of the leading edges or the trailing edges.

(42) The ceiling fan of claim any preceding clause, wherein the tip housing extends along at least a portion of both the leading edges and trailing edges.

(43) The ceiling fan of claim any preceding clause, wherein at least a portion of the tip housing is received in the second interior.

(44) The ceiling fan of claim any preceding clause, wherein the stationary outer housing is radially spaced from the stationary inner housing to form a radial gap that defines the flow channel.

(45) The ceiling fan of claim any preceding clause, wherein the rotor channel and tip channel at least partially overlap axially.

(46) The ceiling fan of claim any preceding clause, further comprising struts coupling the stationary outer housing to the stationary inner housing.

(47) The ceiling fan of claim any preceding clause, wherein the struts are vanes.

(48) The ceiling fan of claim any preceding clause, wherein the stationary inner housing has a contoured first surface forming at least a portion of the flow channel.

(49) The ceiling fan of claim any preceding clause, wherein the stationary outer housing has a contoured second surface forming at least a portion of the flow channel.

(50) The ceiling fan of claim any preceding clause, wherein the tip channel is located within the contoured second surface.

(51) The ceiling fan of claim any preceding clause, wherein the tip housing closes the tip channel.

(52) The ceiling fan of claim any preceding clause, wherein the tip housing comprises an outer surface following a contour of the contoured second surface.

(53) The ceiling fan of claim any preceding clause, wherein the tip housing closes the tip channel.

(54) The ceiling fan of claim any preceding clause, further comprising an air splitter fluidly coupled to the flow channel.

(55) The ceiling fan of claim any preceding clause, wherein the air splitter is located at an inlet to the flow channel.

(56) The ceiling fan of claim any preceding clause, further comprises a light carried by at least one of the stationary outer housing or the stationary inner housing.

(57) The ceiling fan of claim any preceding clause, wherein the light is a plurality of light emitting diodes.

(58) A ceiling fan comprising: a stationary housing assembly defining a first interior and having a

circumferential rotor channel; a motor assembly located within the first interior and having a rotor aligned with the rotor channel; a plurality of blades extending between corresponding roots and tips, with the roots coupled to the rotor; and a tip housing coupled to the blade tips.

(59) The ceiling fan of claim any preceding clause, wherein the tip housing comprises a ring.

(60) The ceiling fan of claim any preceding clause, wherein the ring has an inner surface facing the stationary housing and defining a flow channel therebetween.

(61) The ceiling fan of claim any preceding clause, further comprises an air splitter fluidly coupled to the flow channel.

(62) The ceiling fan of claim any preceding clause, wherein the air splitter is located at an inlet to the flow channel.

(63) The ceiling fan of claim any preceding clause, further comprises a light carried by the stationary housing.

Claims

1. A ceiling fan comprising: a stationary inner housing defining a first interior and having a circumferential rotor channel; a stationary outer housing, circumscribing at least a portion of the stationary inner housing to define a flow channel therebetween, and defining a second interior having a circumferential tip channel; a motor assembly located within the first interior and having a rotor aligned with the rotor channel; a plurality of blades extending between corresponding roots and tips, with the roots coupled to the rotor; and a tip housing coupled to the tips of the blades and at least partially received within the tip channel.
2. The ceiling fan of claim 1, wherein the tip housing extends fully along the tips.
3. The ceiling fan of claim 2, wherein the blades have corresponding leading edges and trailing edges extending from the roots to the tips, and the tip housing extends along at least a portion of the leading edges or the trailing edges.
4. The ceiling fan of claim 3, wherein the tip housing extends along at least a portion of both the leading edges and trailing edges.
5. The ceiling fan of claim 1, wherein at least a portion of the tip housing is received in the second interior.
6. The ceiling fan of claim 1, wherein the stationary outer housing is radially spaced from the stationary inner housing to form a radial gap that defines the flow channel.
7. The ceiling fan of claim 6, wherein the rotor channel and tip channel at least partially overlap axially.
8. The ceiling fan of claim 7, further comprising struts coupling the stationary outer housing to the stationary inner housing.
9. The ceiling fan of claim 8, wherein the struts are vanes.
10. The ceiling fan of claim 1, wherein the stationary inner housing has a contoured first surface forming at least a portion of the flow channel.
11. The ceiling fan of claim 10, wherein the stationary outer housing has a contoured second surface forming at least a portion of the flow channel.
12. The ceiling fan of claim 11, wherein the tip channel is located within the contoured second surface.
13. The ceiling fan of claim 12, wherein the tip housing closes the tip channel.
14. The ceiling fan of claim 13, wherein the tip housing comprises an outer surface following a contour of the contoured second surface.
15. The ceiling fan of claim 1, wherein the tip housing closes the tip channel.
16. The ceiling fan of claim 1, further comprising an air splitter fluidly coupled to the flow channel.
17. The ceiling fan of claim 16, wherein the air splitter is located at an inlet to the flow channel.
18. The ceiling fan of claim 1, further comprises a light carried by at least one of the stationary

outer housing or the stationary inner housing.

19. The ceiling fan of claim 18, wherein the light is a plurality of light emitting diodes.
